Do Firms Truly Outperform the Market? A Bayesian Perspective

Daniel Fu and Jack Wong

Key Research Questions

- Do value firms outperformance the SPY return benchmark
- What is each firm's probability of beating the SPY?
- Do value ratios: P/E ratio or dividend yield explain differences?
- Limited annual observations per firm motivates Bayesian framework

Data and Variables

Data

Used all S&P 500 stock Price Return, P/E, Dividend Yield Metrics that has last 10 year data

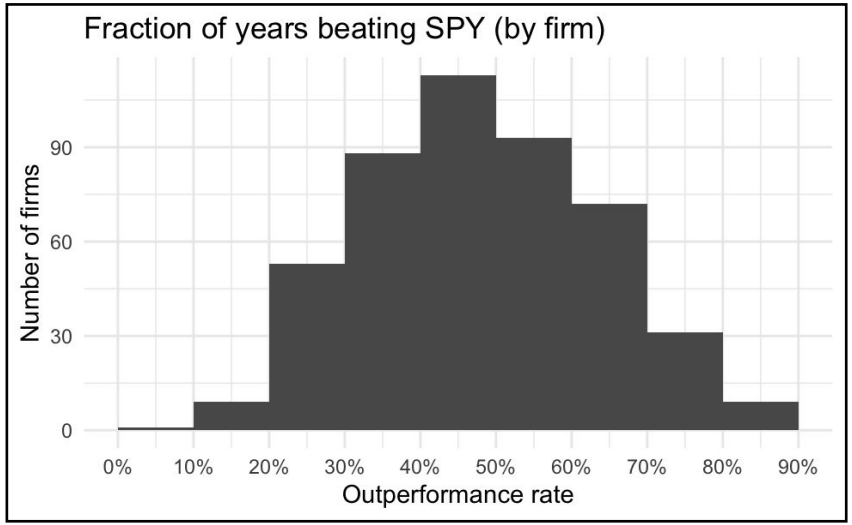
→

Variables

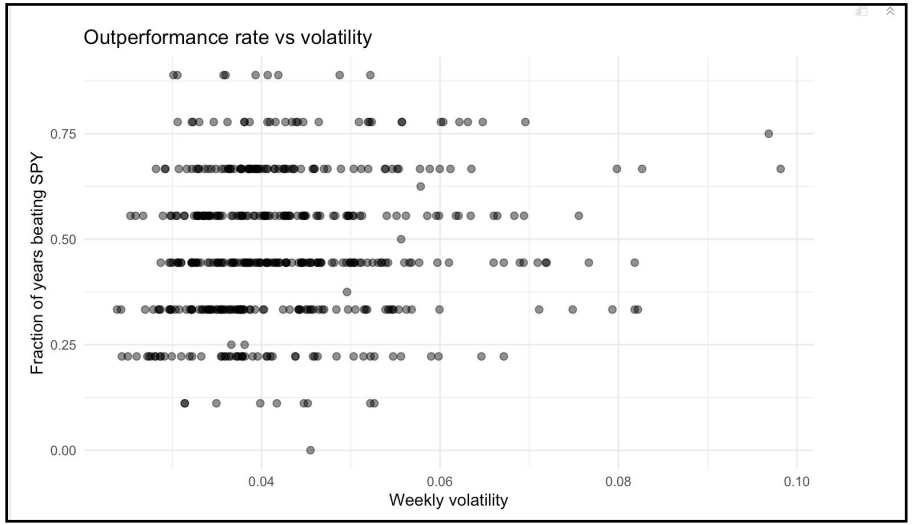
- Weekly price data is compounded into weekly, then annual returns
- Volatility is the standard deviation of weekly returns
- Outperformance is the firms fraction of years beating the SPY benchmark (1 or 0)
- P/E & Dividend Yield measured at start of year to avoid look ahead bias

Exploratory Data Analysis

Skewed towards best firms?

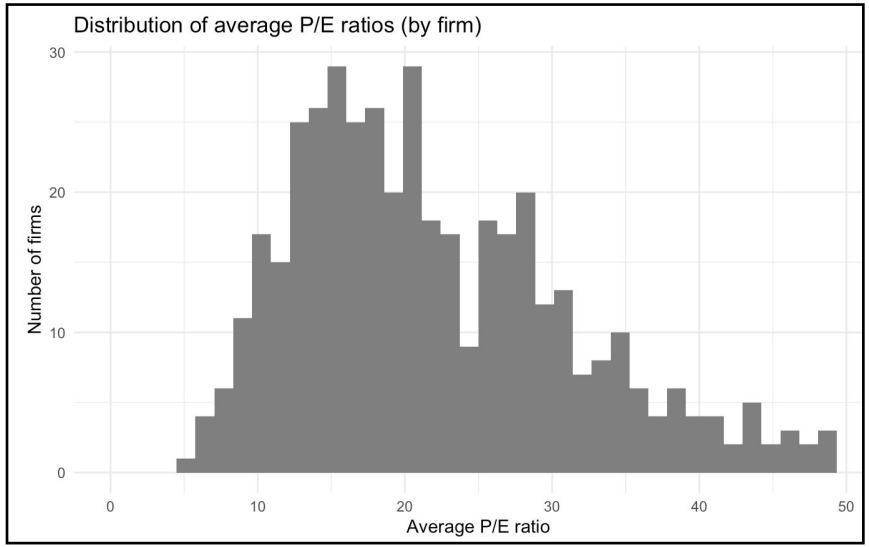


Noise or True Outperformance?

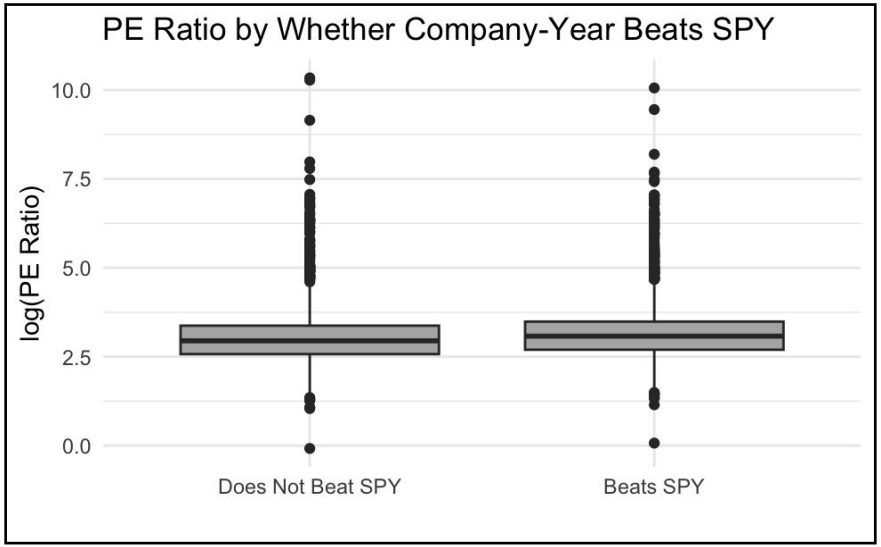


Exploratory Data Analysis (Impact of P/E Ratio)

Distribution of Avg P/E Ratio



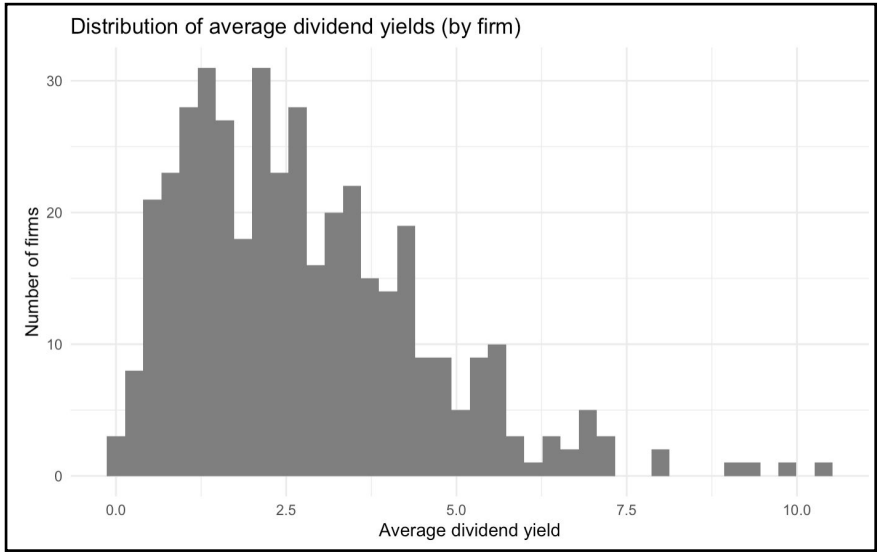
Slightly Higher P/E Ratios Outperforms



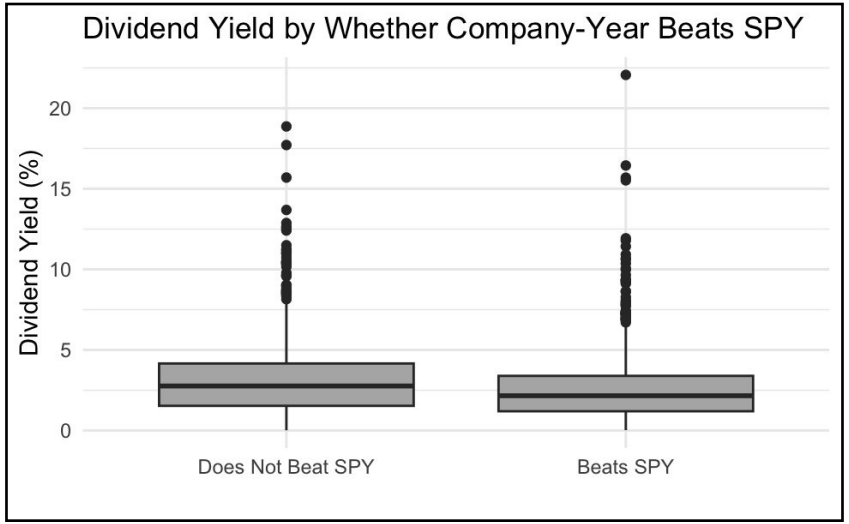
*(Not Pictured) Industry differences show similar result or no difference

Exploratory Data Analysis (Impact of Div Yield)

Distribution of Avg Div Yield



Lower Div Yields Outperforms?



*(Not Pictured) Industry differences show similar result or no difference

Frequentist Analysis

Baseline Frequentist Logistic Model

Given Y ~ Bernoulli where p is the probability that firm *i* outperformed SPY in year *t*

$$logit(p_{it}) = \beta_0 + \beta_1 log(PE_{it}) + \beta_2 DivYield_{it}$$

Table 2: Frequentist Baseline Logistic Model Summary

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-0.18	0.19	-0.92	0.36
log_pe	-0.03	0.06	-0.55	0.58
div_yield	0.15	0.02	7.68	0.00

Statistically significant:
Higher div yield = higher probability of beating SPY

Limitation: Model treats all firms the same aside from these two predictors

Frequentist Logistic Model: Mixed Effect

Model Specification:

$$logit(p_{it}) = \beta_0 + \beta_1 log(PE_{it}) + \beta_2 DivYield_{it} + \beta_{0i} + \beta_{1i} log(PE_{it}) + \beta_{2i} DivYield_{it}$$

Firms can have different baseline probabilities of beating SPY (random intercepts)

And the effect of log(P/E) and dividend yield can vary by firm (random slopes)

Results:

Table 3: Frequentist Mixed Effect Model for Firm

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-0.25	0.20	-1.22	0.22
log_pe	-0.02	0.06	-0.38	0.70
div_yield	0.17	0.02	8.06	0.00

Dividend yield is the most consistent predictor of outperformance; valuation multiple

Table 4: Random Effect Covariance Matrix

	grp	var1	var2	vcov	sdcor
1	Ticker	(Intercept)		0.29	0.54
2	Ticker	log_pe		0.01	0.11
3	Ticker	div_yield		0.01	0.09
4	Ticker	(Intercept)	log_pe	-0.06	-1.00
5	Ticker	(Intercept)	div_yield	-0.05	-1.00
6	Ticker	log_pe	div_yield	0.01	1.00

Most of the differences come from firms having different starting levels of outperformance, not from the predictors affecting firms in very different ways.

Limitations of Mixed Effect:
Random slope correlations can become unstable / near-boundary

Sector FE gives similar predictive performance with much more stability

Frequentist Logistic Model: Sector Fixed Effects

Why use this model?

- ✗ Baseline model too simplistic
- ✗ Overparameterization when modeling each firm effect
- ✓ "Middle Ground" by modeling per sector effect

Model Specification:

$$logit(p_{it}) = \beta_0 + \beta_1 log(PE_{it}) + \beta_2 DivYield_{it} + \eta_1 + \eta_2 + \cdots + \eta_{10}$$

Table 5: Frequentist Sector FE Model Summary

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	0.51	0.31	1.65	0.10
log_pe	-0.16	0.06	-2.49	0.01
div_yield	0.10	0.02	4.37	0.00
gics_sector_nameConsumer Discretionary	-0.26	0.25	-1.05	0.29
gics_sector_nameConsumer Staples	0.26	0.25	1.08	0.28
gics_sector_nameEnergy	0.03	0.26	0.11	0.91
gics_sector_nameFinancials	-0.43	0.23	-1.85	0.06
gics_sector_nameHealth Care	0.13	0.24	0.52	0.61
gics_sector_nameIndustrials	-0.31	0.23	-1.35	0.18
gics_sector_nameInformation Technology	-0.67	0.24	-2.75	0.01
gics_sector_nameMaterials	0.01	0.26	0.04	0.97
gics_sector_nameReal Estate	0.40	0.27	1.49	0.14
gics_sector_nameUtilities	-0.13	0.25	-0.51	0.61

Key result is that dividend yield remains **strongly positive**, and once we control for sector, log(P/E) becomes **significantly negative.**

Some sectors (like IT) have a lower baseline probability of beating SPY relative to Communication Services

Sector Fixed Effects: Model Diagnostics

Figure 8: Component Residual Plots

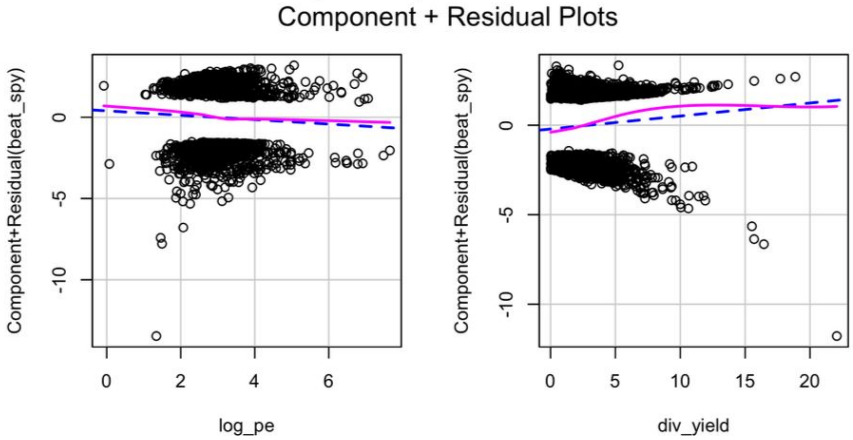


Table 9: VIF Results

	GVIF	Df	GVIF^(1/(2*Df))
log_pe	1.48	1.00	1.22
div_yield	1.52	1.00	1.23
gics_sector_name	1.65	10.00	1.03

Linearity assumptions reasonably hold

No multicollinearity

Sector Fixed Effects: Model Diagnostics

Figure 9: Cook's Distances

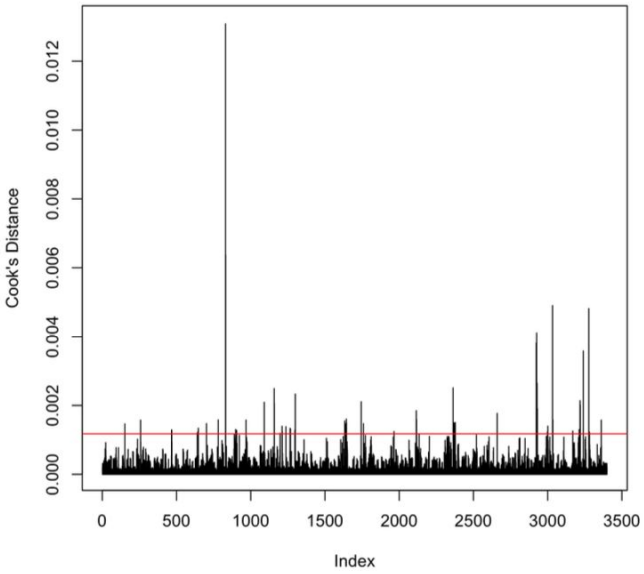


Table 10: Frequentist Firm Fixed Effects w/o Influential Points

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	1.25	0.38	3.28	0.00
log_pe	-0.22	0.07	-2.90	0.00
div_yield	0.17	0.03	6.44	0.00
gics_sector_nameConsumer Discretionary	-1.02	0.30	-3.41	0.00
gics_sector_nameConsumer Staples	-0.49	0.30	-1.65	0.10
gics_sector_nameEnergy	-0.61	0.32	-1.91	0.06
gics_sector_nameFinancials	-1.20	0.29	-4.20	0.00
gics_sector_nameHealth Care	-0.58	0.30	-1.95	0.05
gics_sector_nameIndustrials	-1.03	0.29	-3.62	0.00
gics_sector_nameInformation Technology	-1.42	0.30	-4.77	0.00
gics_sector_nameMaterials	-0.69	0.31	-2.24	0.03
gics_sector_nameReal Estate	-0.36	0.32	-1.13	0.26
gics_sector_nameUtilities	-1.00	0.30	-3.28	0.00

PE and dividend yield effects remain consistent
Opt against removing influential points for future modelling

Frequentist Model Comparison

Table 6: AIC Comparison

	df	AIC
Sector Fixed Effects	13.00	4579.02
Baseline Logistic	3.00	4624.37
Firm Mixed Effects	9.00	4632.86

Table 8: AUC from 10-fold CV

	model	auc
1	Sector Fixed Effects	0.61
2	Firm Mixed Effects	0.59
3	Baseline Logistic	0.59

We observe that the sector fixed effects model **has lower AIC** and **marginally higher AUC**

Since we want to produce a Bayesian model for predictive purposes, we proceed with the **sector fixed effects** model

Bayesian Analysis

Bayesian Analysis: Baseline Logistic Model

Table 2: Frequentist Baseline Logistic Model Summary

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-0.18	0.19	-0.92	0.36
log_pe	-0.03	0.06	-0.55	0.58
div_yield	0.15	0.02	7.68	0.00

Consistent with Frequentist:
Higher div yield = higher probability of beating SPY,

P/E coefficient is insignificant

Baseline Logistic Model – MCMC Convergence

Figure 10: Trace Plots for Baseline Bayesian Model

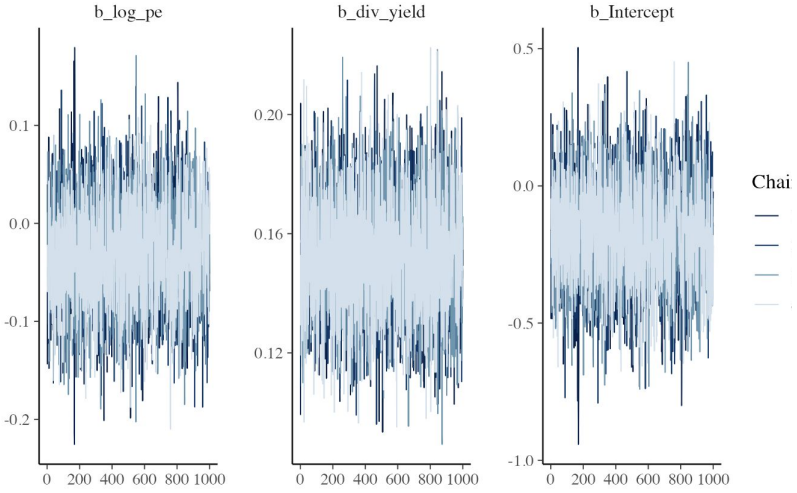
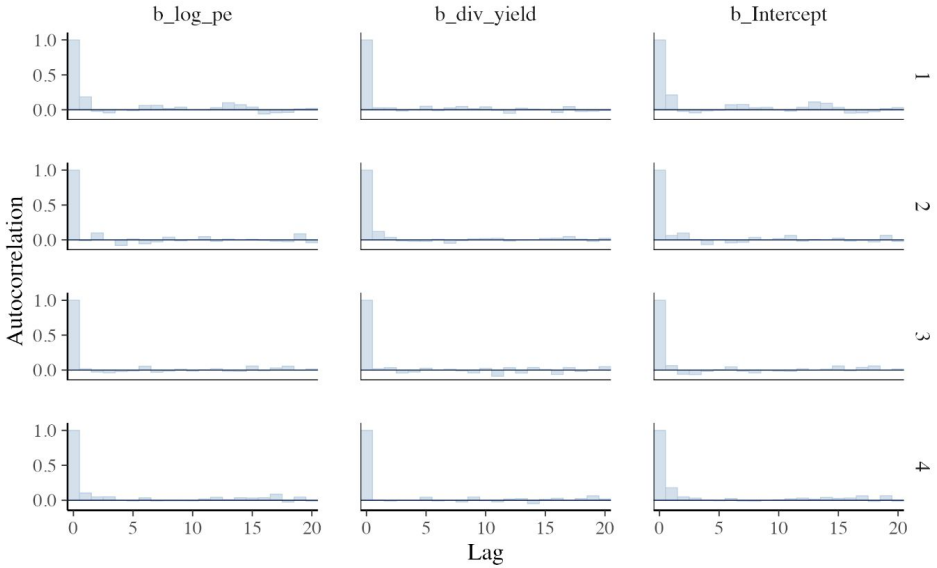


Figure 11: ACF Plots for Baseline Bayesian Model



Bayesian Analysis: Sector Fixed Effects Logistic Model

Table 13: Summary of Bayesian Sector FE Model

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	0.51	0.31	-0.10	1.14
log_pe	-0.16	0.07	-0.29	-0.03
div_yield	0.10	0.02	0.06	0.15
gics_sector_nameConsumerDiscretionary	-0.27	0.25	-0.77	0.22
gics_sector_nameConsumerStaples	0.26	0.25	-0.23	0.75
gics_sector_nameEnergy	0.02	0.27	-0.52	0.53
gics_sector_nameFinancials	-0.43	0.23	-0.90	0.00
gics_sector_nameHealthCare	0.12	0.25	-0.38	0.59
gics_sector_nameIndustrials	-0.32	0.23	-0.79	0.12
gics_sector_nameInformationTechnology	-0.68	0.25	-1.17	-0.21
gics_sector_nameMaterials	0.00	0.26	-0.51	0.53
gics_sector_nameRealEstate	0.39	0.27	-0.16	0.91
gics_sector_nameUtilities	-0.14	0.25	-0.64	0.34

The frequentist and Bayesian sector FE results are consistent

(1) log(P/E) is negative
(2) Dividend yield is positive
(3) IT has a lower baseline outperformance probability

Sector FE Logistic Model - MCMC Convergence

Figure 12: Trace Plots for Sector FE Bayesian Model

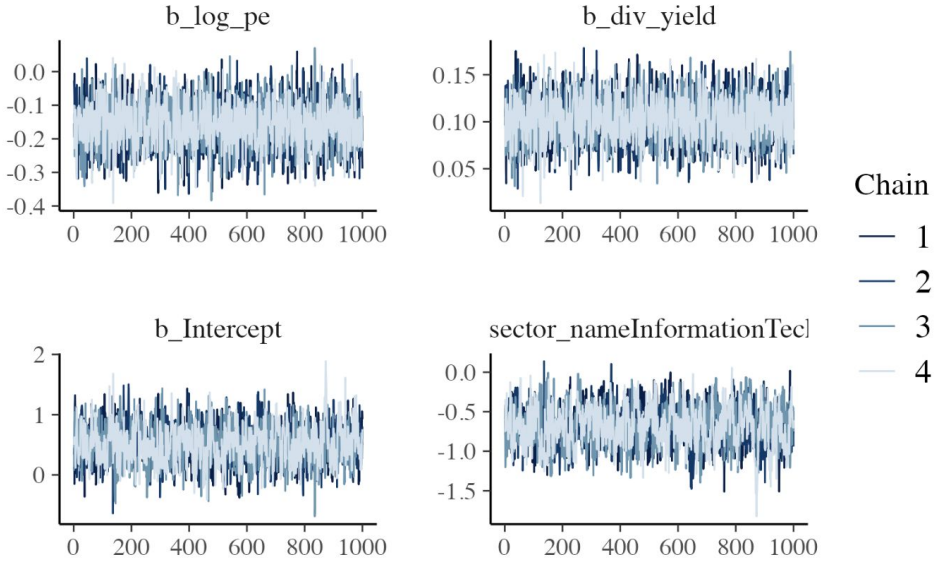
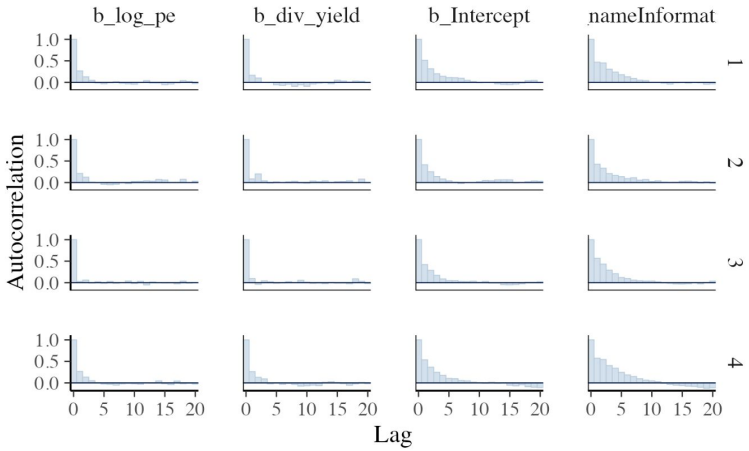
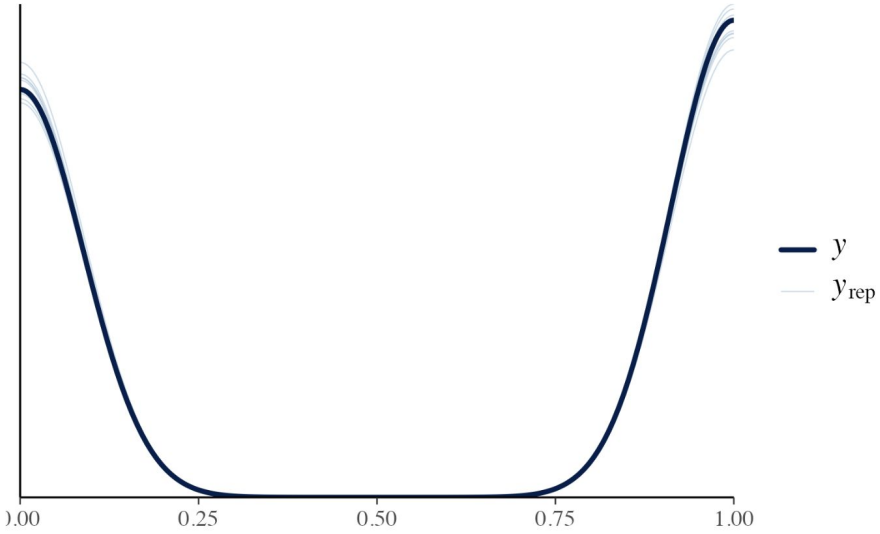


Figure 13: ACF Plots for Sector FE Bayesian Model



Posterior Predictive Check

Figure 14: PPC for Sector FE Model (default prior)



Simulated data generated from the posterior closely matches the observed distribution

Bayesian Model Comparison

Table 14: Leave-one-out Comparison of Bayesian Models

	elpd_diff	se_diff	elpd_loo	se_elpd_loo	p_loo	se_p_loo	looic	se_looic
bayes_fe	0.00	0.00	-2289.83	12.82	13.50	0.28	4579.67	25.63
bayes_baseline	-22.44	8.04	-2312.28	10.22	3.23	0.18	4624.56	20.44

ELDP indicates that the sector fixed effects model has significantly better predictive performance than the baseline model.

Sensitivity Analysis

- Up to this point, models were fit with default brm priors
- Test sensitivity of model using two new sets of priors

Slopes: t(3, 0, 2.5)
Intercept: t(3, 0, 5)

Slopes: N(0,1)
Intercept: N(0,2)

Sensitivity Analysis

Table 14: Sector FE model using Weakly Informative Priors

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.51	0.31	-0.07	1.12	1.00	1058.89	1521.78
log_pe	-0.16	0.06	-0.29	-0.04	1.00	2424.54	2642.91
div_yield	0.10	0.02	0.06	0.15	1.00	2249.90	2325.14
gics_sector_nameConsumerDiscretionary	-0.25	0.24	-0.73	0.21	1.00	798.26	1187.32
gics_sector_nameConsumerStaples	0.27	0.24	-0.22	0.73	1.00	765.05	1314.17
gics_sector_nameEnergy	0.04	0.27	-0.48	0.55	1.00	882.46	1617.17
gics_sector_nameFinancials	-0.42	0.23	-0.87	0.01	1.00	721.29	1084.08
gics_sector_nameHealthCare	0.13	0.24	-0.34	0.59	1.00	766.88	1139.73
gics_sector_nameIndustrials	-0.31	0.23	-0.75	0.12	1.00	693.95	1205.59
gics_sector_nameInformationTechnology	-0.66	0.24	-1.15	-0.21	1.00	777.80	1181.99
gics_sector_nameMaterials	0.02	0.26	-0.48	0.50	1.00	816.27	1434.06
gics_sector_nameRealEstate	0.40	0.27	-0.11	0.94	1.00	849.89	1613.50
gics_sector_nameUtilities	-0.12	0.25	-0.61	0.36	1.00	803.12	1359.53

Table 15: Sector FE model using Weakly Informative Priors

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.48	0.29	-0.08	1.04	1.00	1276.40	1792.04
log_pe	-0.16	0.06	-0.28	-0.03	1.00	2419.47	2574.31
div_yield	0.10	0.02	0.06	0.15	1.00	2468.07	2717.92
gics_sector_nameConsumerDiscretionary	-0.23	0.22	-0.66	0.19	1.00	1034.60	1745.07
gics_sector_nameConsumerStaples	0.28	0.22	-0.14	0.69	1.00	967.56	1458.58
gics_sector_nameEnergy	0.05	0.23	-0.40	0.51	1.00	1128.68	1732.61
gics_sector_nameFinancials	-0.40	0.20	-0.79	-0.01	1.00	876.13	1537.23
gics_sector_nameHealthCare	0.15	0.21	-0.27	0.55	1.00	974.85	1526.90
gics_sector_nameIndustrials	-0.29	0.20	-0.67	0.09	1.00	879.02	1515.31
gics_sector_nameInformationTechnology	-0.64	0.22	-1.06	-0.23	1.00	1006.10	1594.49
gics_sector_nameMaterials	0.03	0.23	-0.41	0.47	1.00	1118.07	1536.79
gics_sector_nameRealEstate	0.42	0.24	-0.06	0.89	1.00	1110.45	1849.02
gics_sector_nameUtilities	-0.11	0.22	-0.53	0.32	1.00	1050.07	1605.28

Consistent results with default prior model and each other

(1) log(P/E) is negative
(2) Dividend yield is positive
(3) IT has a lower baseline outperformance probability

Sensitivity Analysis

Figure 15: PPC of Sector FE Model w/ weakly informative priors

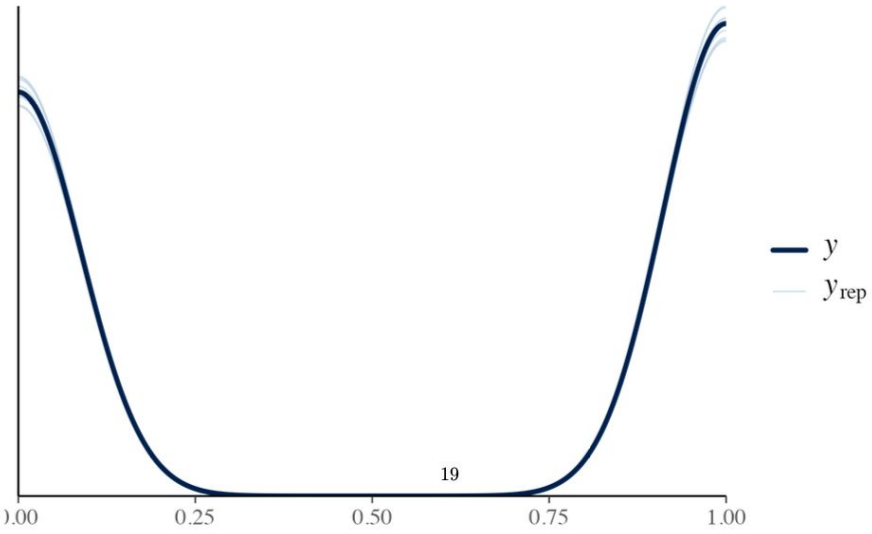
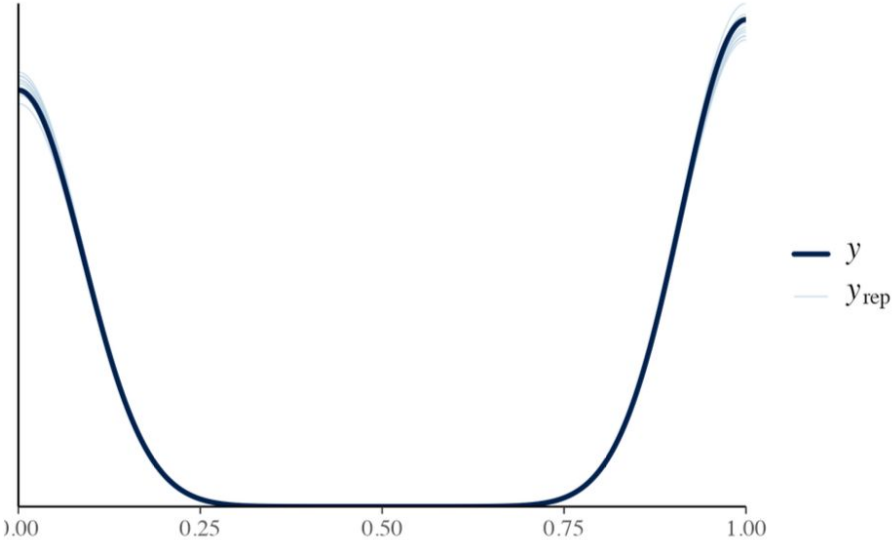
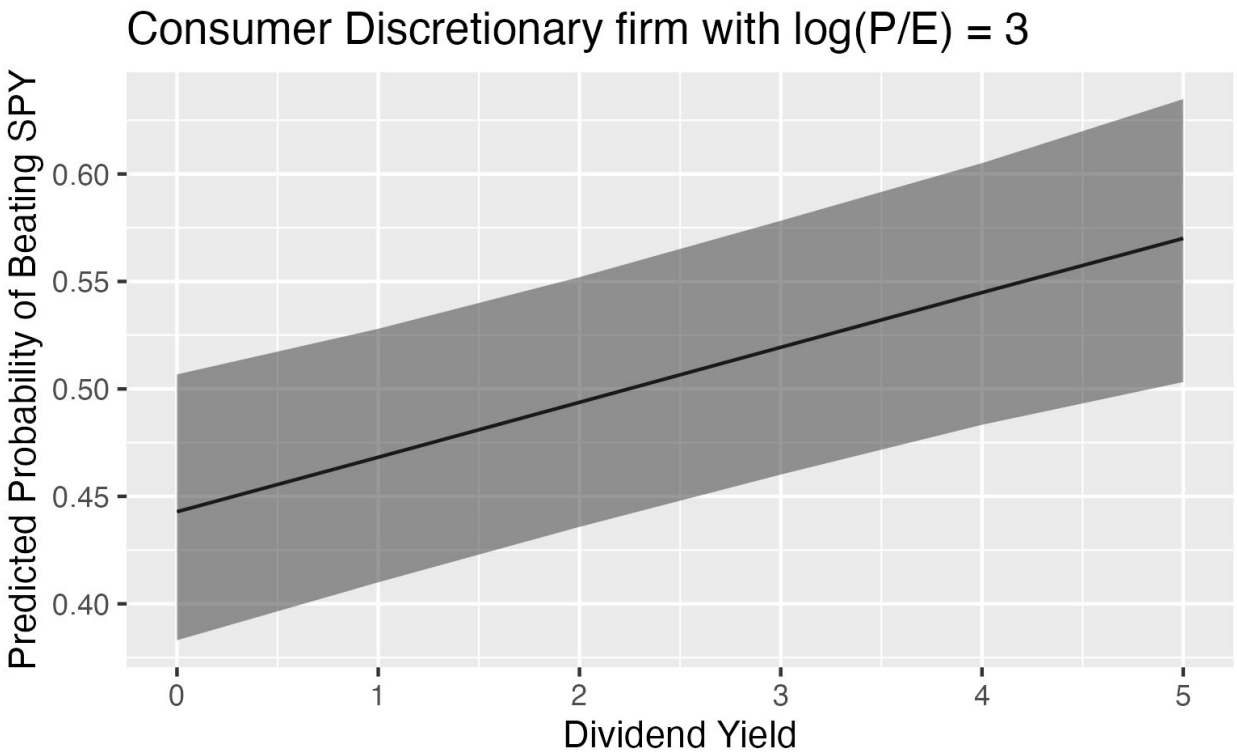


Figure 16: PPC of Sector FE Model w/ slightly regularizing priors



Prediction


Consumer Discretionary firm with log(P/E) = 3
Predicted Probability of Beating SPY
0.60
0.55
0.50
0.45
0.40
0
1
2
3
4
5
Dividend Yield

Discussion

- Remove influential points and refitting model
- Explore different mixed models
  - Random effects by sector
- Add additional predictors